

Exact Finite Dimer Random Sequential Adsorption on Paths and Cycles

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Abstract

We give a finite-size solution of dimer random sequential adsorption on a path and a simple cycle, viewed as a uniformly random greedy edge order. Conditioning on the first edge closes a convolution for the probability generating polynomials and a Riccati ordinary generating function. It gives an exact mean and the cycle identity $G_n(z) = zF_{n-2}(z)$.

1 Model, ownership, and claim boundary

Let P_n have vertices $1, \dots, n$ and edges $e_i = \{i, i+1\}$, with P_0 empty. Assign iid continuous priorities to the edges and inspect them in increasing order. An edge is accepted exactly when both endpoints are currently unmatched. Equivalently, scan a uniformly random permutation of the labeled edges. Let M_n be the number accepted and write

$$F_n(z) = \mathbb{E}z^{M_n}.$$

For the simple labeled cycle C_n , $n \geq 3$, write the corresponding count and PGF as K_n and $G_n(z)$.

The terminal edge set is a jammed *maximal* matching: no further edge can be added. It need not be a *maximum* matching. This distinction is part of every statement below. Continuous priorities merely generate a uniform order; a discrete-priority model needs a separate tie-breaking convention.

One-dimensional blocking goes back at least to Flory's polymer calculation [1]; Evans reviews random and cooperative sequential adsorption [2]. Penrose gives rigorous general jamming-limit context [3], while Dyer and Frieze place randomized greedy matching in its algorithmic lineage [4]. We reconstruct an exact finite theorem and integrity chain. We do not claim invention or literature priority for the model, the limiting density, or random greedy matching.

2 The finite theorem

Theorem 1 (Path/cycle PGFs, moments, support, and boundary effect). *Set $F_0 = F_1 = 1$. For every $n \geq 2$,*

$$(n-1)F_n(z) = z \sum_{a+b=n-2} F_a(z)F_b(z). \quad (1)$$

Thus the formal OGF $F(x, z) = \sum_{n \geq 0} F_n(z)x^n$ is the unique formal solution with $[x^0]F = [x^1]F = 1$ of

$$F_x = \frac{F-1}{x} + zx F^2, \quad xF_x - F + 1 = zx^2 F^2. \quad (2)$$

Put $H_r(x) = \partial_z^r F(x, z)|_{z=1}$. Then $H_0 = (1-x)^{-1}$, and for every $r \geq 1$, $H_r(0) = 0$ and

$$\begin{aligned} H'_r - \left(\frac{1}{x} + \frac{2x}{1-x} \right) H_r &= x \sum_{j=1}^{r-1} \binom{r}{j} H_j H_{r-j} \\ &\quad + rx \sum_{j=0}^{r-1} \binom{r-1}{j} H_j H_{r-1-j}. \end{aligned} \quad (3)$$

This triangular system determines every factorial moment. In particular, for $n \geq 2$,

$$\mathbb{E}M_n = \sum_{j=1}^{n-1} (n-j) \frac{(-1)^{j+1} 2^{j-1}}{j!}. \quad (4)$$

If $a = e^{-2}$ and $\alpha = (1-a)/2$, then

$$\mathbb{E}M_n = \alpha n - a + o(1), \quad \text{Var}(M_n) = a^2 n + 2a^2 + o(1). \quad (5)$$

The support is $\{0\}$ for $n = 0, 1$. For $n \geq 2$,

$$\text{supp}(M_n) = \left\{ \left\lceil \frac{n-1}{3} \right\rceil, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\}. \quad (6)$$

For every simple cycle $n \geq 3$,

$$G_n(z) = zF_{n-2}(z), \quad \text{supp}(K_n) = \left\{ \left\lceil \frac{n}{3} \right\rceil, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\}. \quad (7)$$

Consequently

$$\mathbb{E}K_n = 1 + \mathbb{E}M_{n-2} = \alpha n + o(1), \quad (8)$$

$$\mathbb{E}K_n - \mathbb{E}M_n = e^{-2} + o(1), \quad (9)$$

$$\text{Var}(K_n) = \text{Var}(M_{n-2}) = e^{-4}n + o(1). \quad (10)$$

Both expected occupied-vertex fractions tend to $1 - e^{-2}$.

3 First-edge decomposition and the Riccati OGF

The first edge in any order is accepted. Conditional on e_i being first, its endpoints $i, i+1$ are blocked and the eligible residual vertices split into P_{i-1} and P_{n-i-1} . Given this event, the relative orders induced on those two disjoint edge sets are independent and uniform. Their interleaving has no effect because the vertex sets are disjoint. Therefore

$$M_n \stackrel{d}{=} 1 + M_{i-1}^{(L)} + M_{n-i-1}^{(R)} \quad \text{given } e_i \text{ first,}$$

where the two variables on the right are independent. Averaging over the $n-1$ equally likely choices proves (1). Multiplication by x^{n-1} and summation over $n \geq 2$ gives

$$F_x - \frac{F-1}{x} = zxF^2,$$

which is (2). This is a formal-power-series derivation, so it does not presume analytic convergence in two variables.

At $z = 1$, normalization gives $F_n(1) = 1$, hence $F(x, 1) = (1-x)^{-1}$. Differentiate zF^2 exactly r times:

$$\partial_z^r (zF^2)|_{z=1} = \sum_{j=0}^r \binom{r}{j} H_j H_{r-j} + r \sum_{j=0}^{r-1} \binom{r-1}{j} H_j H_{r-1-j}.$$

The $j = 0, r$ terms in the first sum are $2H_0H_r$. Moving them to the left proves the all- r equation (3); all other indices are strictly below r .

For $r = 1$, the equation is

$$H_1' - \left(\frac{1}{x} + \frac{2x}{1-x} \right) H_1 = \frac{x}{(1-x)^2},$$

and its formal solution regular at zero is

$$H_1(x) = \frac{x(1 - e^{-2x})}{2(1-x)^2}. \quad (11)$$

Expanding the exponential and extracting $[x^n]$ proves (4).

References

- [1] P. J. Flory, Intramolecular reaction between neighboring substituents of vinyl polymers, *J. Amer. Chem. Soc.* **61**(6), 1518–1521 (1939), [doi:10.1021/ja01875a053](https://doi.org/10.1021/ja01875a053).
- [2] J. W. Evans, Random and cooperative sequential adsorption, *Rev. Mod. Phys.* **65**(4), 1281–1329 (1993), [doi:10.1103/RevModPhys.65.1281](https://doi.org/10.1103/RevModPhys.65.1281).
- [3] M. D. Penrose, Random parking, sequential adsorption, and the jamming limit, *Commun. Math. Phys.* **218**(1), 153–176 (2001), [doi:10.1007/s002200100387](https://doi.org/10.1007/s002200100387).
- [4] M. Dyer and A. Frieze, Randomized greedy matching, *Random Structures Algorithms* **2**(1), 29–45 (1991), [doi:10.1002/rsa.3240020104](https://doi.org/10.1002/rsa.3240020104).